WestRock

PI tag Standard Document

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1. Executive Summary

The primary goal of this document is to provide a reference for building PI tags, configuring sampling, and compression for the optimal data quality, resolution, and resource loading of the IT equipment. This document will provide guidance to engineering and plant resources in creating new PI tags and properly configuring PI software. This document contains several scenarios based on variable services. It will also show best practice to use to edit existing PI tags for improved data quality.

1.1. Benefits

By applying the standards discussed in this document the following benefits can be achieved:

1.1.1. Corporate Business Leaders

- Improved visibility of production capability/capacity through real-time data.
- Having standard best practice for data compression will permit accurate benchmarking of similar variables.

1.1.2. Plant Operations/Leaders

- Improve the stability of plant operations by minimizing data loss due to improper configuration.
- Improve supply chain management through plant floor visibility.
- Plant-to-business data integration allowing for the visibility of key performance indicators.
- Reduce training costs for network support resources by choosing standard solutions and encouraging plant to plant interchangeability.
- Improvement of plant floor/process visibility to track and improve OEE.

1.1.3. IT Stakeholders

- Improved performance through standardized infrastructure.
- Improve network maintainability using standard OSISoft interface configuration.

1.2. Glossary

The following are common abbreviations utilized in this document:

Abbreviation	Definition
OPC	Open Platform Communications
DCS	Distributed Control System
API	Application programming interface
ADC	Analog Digital Converter
PLC	Programmable Logic Controller
QCS	Quality Control System
SQC	Statistical Quality Control

Table 1: A list of abbreviations used throughout the document.

2. Document Scope

This document discusses standards specifically related to PI tag creation, tag naming, sampling rate, and data compression.

2.1. PI tag creation

PI tag creation is the foundation of organizing data from multiple sources. This document will list the required information necessary to build a PI tag that will represent a variable from multiple sources such as DCS, PLC, QCS, manual data and calculations.

2.2. Tag naming

The scope of this document regarding tag naming will show guidelines on proper naming practices. Considering the history of our mills and the multiple tag naming standards written by mills, this document will describe some modifications necessary to improve tag searching capabilities and reduce duplicate tags.

Considering the process and control systems will be upgraded over time, it is necessary to have PI tag that represent the process variable rather than the physical connection to a DCS/PLC. The objective is to simplify the maintenance of the tag database.

2.3. Sampling rate

To be able to compare similar processes, it is necessary to standardize the sampling rate of the process data. This document will show standard sampling rate of the data. We will cover PI interface configuration with a recommended sampling frequency.

2.4. Data compression

Why is data compression so important? More data does not equal better data quality. Two major faults being observed in current systems are too little and too much data. The objective is to present the users an accurate representation of the process which will allow them to make better operational decisions.

Data compression standard and best practice will be covered in this document. This document will show how to configure a PI tag for best data resolution without overburdening the IT infrastructure. We will cover guidelines on how to “tune” a PI tag for data quality.

3. PI System Data Architecture

3.1. Data Flow

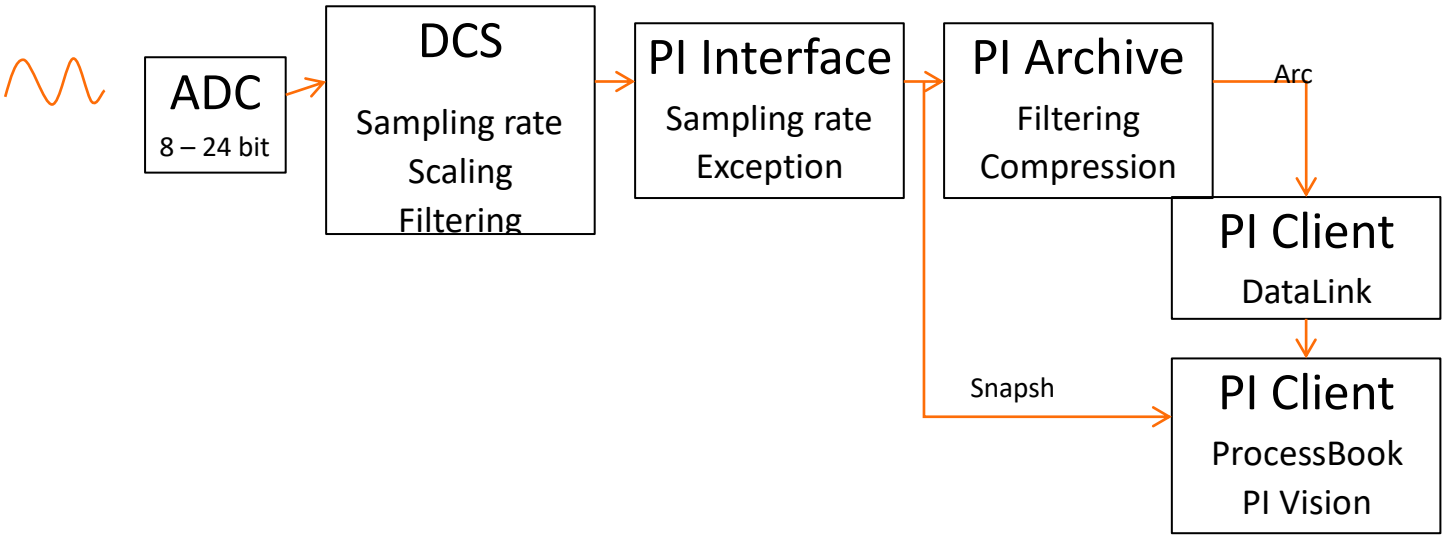


Figure 1: Data Flow Diagram

The above figure shows the data flow from the field instrument to the PI clients. There are many steps where the field variable is manipulated and transformed. It is important to understand what each step does to the data. The objective is to re-create in the PI client an accurate representation of the process.

- The analog to digital converter (ADC) is the first device where the analog variable is converted to digital data for the DCS to control. The resolution of the data is dependent on the resolution of the ADC. It can be from 8 bit to 24 bit.

  As an example, an 8-bit converter has 2^8 = 256 steps and a 16-bit converter has 2^16 = 65,536 steps. The final best resolution of the measured variable will be the calibrated span of the field instrument divided by the number of steps of the ADC (100% / 256 steps = 0.39% per step for an 8-bit ADC).

- Next in the data flow is the DCS. Every manufacturer has different methods to transform the data for the PID control block. Some DCS applies a delta to prevent overloading the controller CPU. This delta reduces the number of values to process. The CPU scan rate may also be a limitation on how much data the PI interface will see. Typically, a DCS scan rate is 1 second.

  Process variable can also be filtered. Filtering can be applied either at the field instrument or at the DCS input module. It is recommended to only use filtering to improve the control and it is also recommended to be applied at the DCS input level over local to field instrument.

- Next in the data flow is the PI interface with the DCS and other data sources. Connectivity to the DCS can be accomplished by 2 main interface types. The older or legacy PI interface were dedicated to a specific DCS manufacturer. The most common interface is now using the OPC protocol.

  The OPC protocol is configured to use an OPC Server to OPC Client architecture. An OPC Server can publish data to multiple OPC client. The PI interface is an OPC client.

The PI OPC interface monitors the published data on a specific scan frequency. If the data exceeds the exception value, then the data is pushed to the PI server. These values are called a “Snapshot”. The snapshot value is also read buy the PI clients such as ProcessBook and PI Vision.

More recent installation will have a firewall in between an upstream and a downstream OPC UA server. The PI OPC interface will connect to the downstream OPC UA server to provide more security to the process control network.

- The final step in the data path is the PI server. The PI server receives snapshot values from the PI interface and processes these values in its’ compression algorithm. The recorded values are known as “archive values”.

  Snapshot values are also sent to connected PI clients such as ProcessBook and PI Vision. The snapshot is defined as the value sent directly from the interface. The snapshot value satisfied the exception configuration for each tag.

3.2. PI Interface

There are mainly two types of PI interfaces with DCS.

- The first one is an API interface connecting to older Unix operating DCS systems such as Bailey and legacy Foxboro to name a few. They are often installed on the DCS equipment and have special configuration requirements. These are older interfaces that will eventually be upgraded to the newer OPC standard.
- The most prevalent PI interface uses the OPC client protocol. This protocol is based on a publisher/subscriber architecture where the publisher reads the data from the DCS and makes it available to subscriber called OPC client.

  The OPC data read by the clients are: the value of the variable, the timestamp of the value and the quality of the data (good, bad or questionable).

  The advantage of this architecture is to limit the number of connections to the DCS from multiple interfaces. The typical scan frequency is like an HMI console or 1 second. The interfaces are then free to read data from the OPC server at whatever frequency and exception.

  Another advantage to OPC is it offers a standard interface configuration regardless of the DCS manufacturer.

3.3. PI Server

The PI server is where data is being archived. It is also where the compression algorithm is applied. The PI server receives snapshot from the PI interface. The snapshots are buffered until the compression algorithm applies compression. The snapshots are then discarded.

The PI server can also perform multiple calculations. These calculations are performed on the incoming snapshots, making them very precise.

3.4. PI Clients

The most popular PI clients are DataLink and ProcessBook. The most recent addition is a web base client application called Vision. These clients connect to the PI server and read archived (compressed) data.

The compressed data is not archived at a specific frequency. The clients read the data value and its timestamp and then draws a straight line between the data points.

ProcessBook has the advantage of being able to display snapshot values. These snapshot values are only temporarily displayed. They will be discarded every time the ProcessBook display is refreshed.

4. PI Interface for OPC DA

4.1. Configuring PI Interface

The configuration of PI interfaces is done using an application called PI ICU (Interface Configuration Utility). This utility connects to the PI server and creates a backup of the interface configuration as a module in the module database.

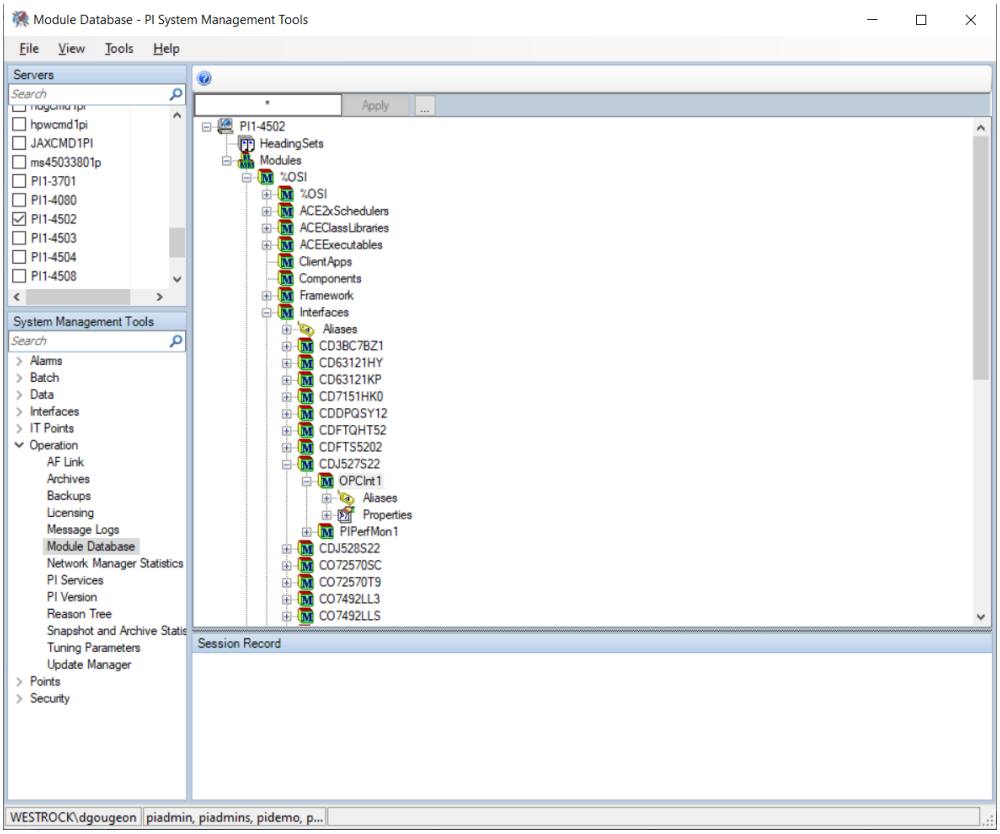


Figure 2: Module database showing interfaces

In figure 2 we can see, using PI SMT (System Management Tool), the complete list of PI interfaces connected to a specific PI server. All interfaces are listed under their location computer name. Multiple interfaces can be installed on a single computer.

To access the interface configuration, you browse to the host interface computer name and right click as shown on figure 3.

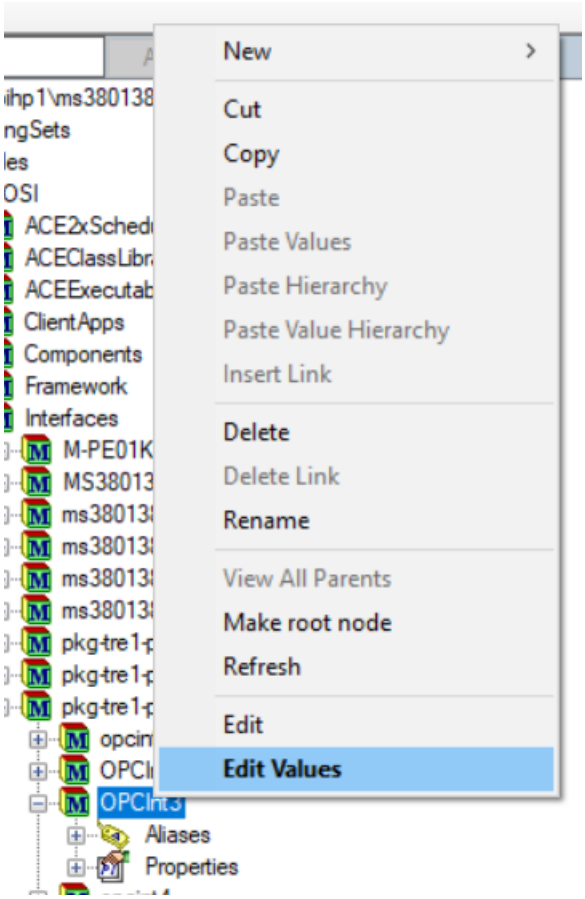


Figure 3: Accessing PI interface configuration details

Once the interface detail configuration parameters window is open (figure 3), then at a minimum, a PI interface requires a point source name (/PS), an ID number (/ID) and scan classes (-/f{xx}). The example below has a point source named “Power”, and ID of 1 and 14 scan classes.

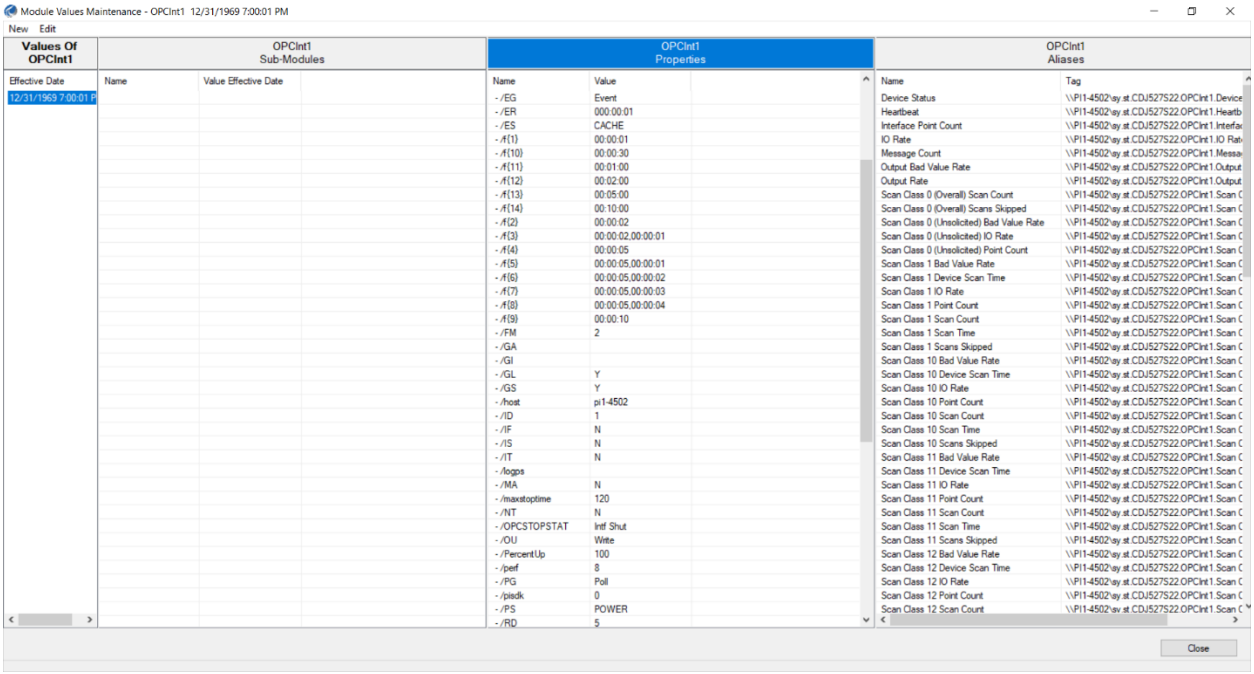


Figure 4: Detail interface configuration window

The point source can be alpha numeric, the ID can only be a numeric and the scan classes represents a time period with this format hh:mm:ss,hh:mm:ss. (ex. -/f{5} = 00:00:05,00:00:01 would be scanned every 5 seconds with an offset of 1 second).

These parameters are critical in creating PI tags as they determine what interface a point is connected to.

4.2. Recommended OPC DA Interface Configuration

It is highly recommended to configure all PI interface to be identical. This will make the PI tag configuration and maintenance considerably simpler.

All interfaces will connect to a specific control system OPC Server. Therefore, the interface configuration can differ regarding connectivity and security parameters.

4.2.1. PointSource and ID Configuration

PI tags connect to an interface using the PointSource and Location 1 parameters. The PointSource can be alpha numeric and is mapped to the -/PS attribute. The Location 1 parameter (-/ID attribute for the interface) can only be numeric. It is possible for PI tags to be mapped to the same PointSource with different Location 1.

4.2.2. Scan Classes Configuration

The scan classes determine at what frequency PI tags are being updated. We recommend that all interfaces be configured with the same scan classes. Generally, OPC server update frequency is similar to a DCS HMI refresh rate. This sampling rate is usually 1 second.

To optimize the PI system data flow and data archiving, experience has shown that a 5 second frequency is more than enough to accurately show the actual process. Faster rate overburdens interface I/O rate and can lead to skipped scans, lower data buffering time, higher PI server disk usage and PI client data retrieval.

These are the recommended scan classes for an OPC DA interface:

-/F{1} = 00:00:01

-/F{2} = 00:00:02

-/F{3} = 00:00:02,00:00:01

-/F{4} = 00:00:05

-/F{5} = 00:00:05,00:00:01

-/F{6} = 00:00:05,00:00:02

-/F{7} = 00:00:05,00:00:03

-/F{8} = 00:00:05,00:00:04

-/F{9} = 00:00:10

-/F{10} = 00:00:30

-/F{11}= 00:01:00

The bulk of the PI tags will be using the 5 second scan classes. Each one of them are offset by 1 second to distribute the load. Faster scan classes should only be used for tags that are pulsed in nature (ex. Refiner plate stepping motor). Slower scan classes (greater than 5 seconds) should be used for tags that do not require fast updates (ex. Loop tuning parameters, production summary, etc.).

The interface scan class number have no reference to the frequency. For maintenance reason and simplicity, scan classes frequencies should always be numbered in an increasing order.

As a best practice, it is recommended to distribute the tags evenly for the five 5 second scan classes. The recommended maximum number of tags should be about 800. However, this is not a hard limit.

It is also recommended to have all tags from a single control loop on the same scan class except for the gain, integral and derivative on a 1 minute scan class.

4.2.3. Polled or Advise Data Handling

The OPC interface has 2 modes of sampling data. The first is polled, where variables are read at the exact time of the scan class. The data will be sent to the PI server even if it did not change since the last scan. The snapshot data will be shown as a step.

The following example show a variable being polled. We can observe the variable stepping like a stair instead of a slope. Each dot represents an archived event.

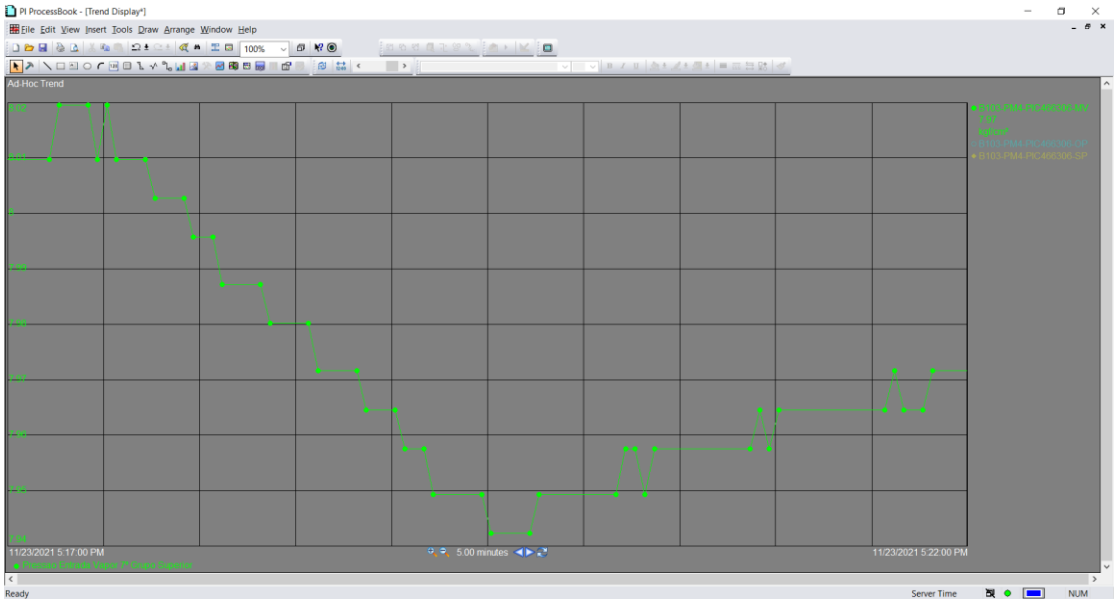


Figure 5: PI display showing polled tag

The preferred configuration would be to be advised. The following display of the same variable shows the difference between a polled tag to the left versus advise to the right of the cursor line.

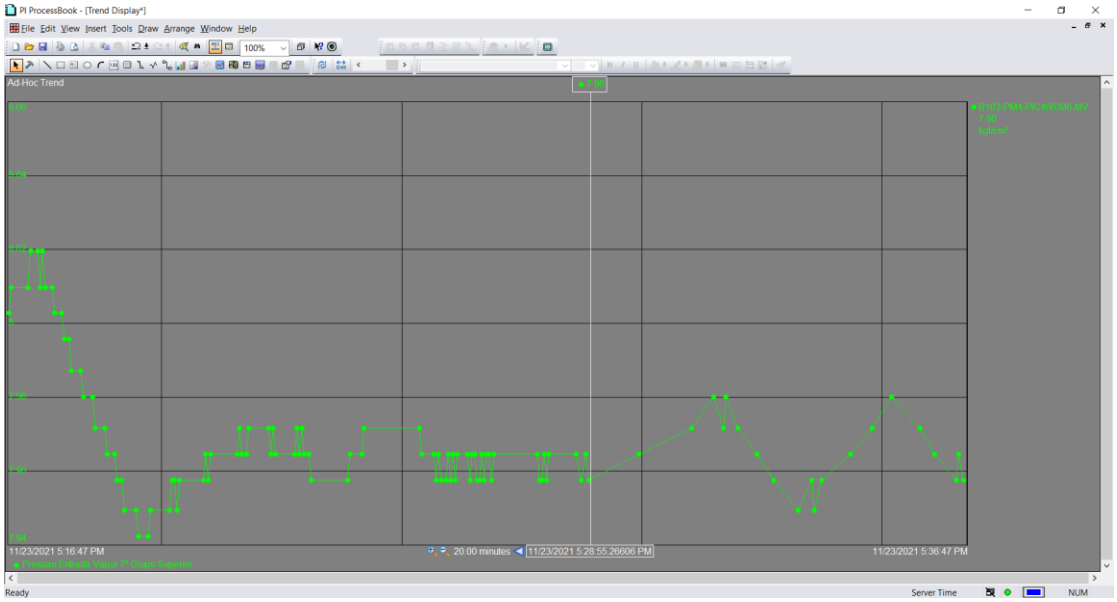


Figure 6: PI display showing polled versus advise tag

Advise tags is recommended as it reduces the amount of data between the PI interface and the PI server. The memory and CPU demand on the interface PC is also optimized. The data being recorded will better represent the process, improve statistical analysis of the data in calculations and the compression algorithm performance will be improved.

Location 3 of a PI tag determines if the data will be polled or advised:

Location 3 = 0 Tag will be polled

Location 3 = 1 Tag will be advised

4.2.4. Timestamp Handling

The OPC protocol is designed to publish the variable value, the timestamp of the variable and the variable quality. The PI interface reads these 3 values and send them to the PI server.

The interface parameter for timestamp handling is named -/TS. It specifies whether time stamps come from the OPC server or from the interface PC when the data arrives. It is an optional parameter, and the default is -/TS = N. The recommended configuration is -/TS = Y.

The timestamp has 4 options:

- -/TS = N This is the default and not the preferred configuration. Setting TS=N means the timestamp applied to the data is coming from the interface PC at the time of the polled or advised read of the data.
- -/TS = Y This is the preferred configuration. This setting means the timestamp will be coming from the DCS when the data value changed. The timestamp returned to the PI server is different depending on if the tag is being polled or is advised.

- For polled tags, the timestamp will match the scan class.
- For advised tags, the timestamp will use the DCS timestamp when the value last changed. The value can change in between scan class read. Considering OPC interface can handle sub-second data, this is important to record first out events. It is also important to view when a process value changed versus a control output change.

- -/TS = A This is to be used only if the OPC server can supply valid timestamp only for advise tags.
- -/TS = U This is not recommended because the interface will not apply time offset.

On interface startup, the time difference between the OPC server and the PI server is measured. Any difference will be adjusted. The offset will be applied to all PI tags using this interface.

4.3. Interface Redundancy

The PI OPC DA interface can be configured for redundancy. There are 3 levels of possible redundancy:

- Single interface to redundant PI servers
- Redundant interfaces to single PI server
- Redundant interfaces to redundant PI servers

This document does not cover interface redundancy because it has no impact on PI tag configuration and data quality.

5. PI tag naming convention

PI tag naming is very important because name is one of the most used search parameters that can be used to find data. It is not a simple matter to standardize tag names due to the history of all the PI servers in our Company.

Depending on the site, tag names have followed many variations. Some of the most common variations are descriptive tag name (ex. 53_PB10_WestSO2CalDriftLimit_PV), instrument tag (ex. B103-PM4-46PT1166-MV), loop number (ex. 11-LIC-1005.PV) and other tag based on acronyms (ex. 09_LL_MOTOR_09_45_2380_INTL_P1_PV).

This tag naming convention is not intended to standardize all existing PI tag names throughout the Company. It will serve as a guideline for each mill with the objective of finding PI tags easier for the end user. This is to serve as a guideline for creating new PI tags.

5.1. Key concept for tag naming

A tag name should have a prefix, a tag name body, and a suffix.

5.1.1. Prefix

The prefix should represent an area of the mill and a major piece of equipment. The area of the mill is often shown as a number that represents the accountings cost number. A major piece of equipment can be a paper machine, a wash line, a digester, etc.

In this example 13-WFB2-OpacityCalGas.PV, the area number is 13 representing the powerhouse and a major piece of equipment is WFB2 for Waste Fuel Boiler #2. It will be easy for the end user to find all tags in the powerhouse by searching 13* or if interested in only WFB2 then the user would use *WFB2* in the search engine.

Each mill has their own prefix standard. Each mill should develop their own standard. New installations and/or upgrades should modify or build the tag database to this standard if practical.

It is highly recommended to use prefix in the PI tag name regardless of the style of tag names used. The prefix should be segregated from the tag name body by either an underscore or a dash. It is important to follow current mill naming practice when using the underscore or dash.

5.1.2. Tag Name Body

The tag name body is **required to be identical** for all the variables of a control loop or motor. This requirement ensures all the variables are kept together. It is not recommended to use the “instrumenttag” as a tag name.

Depending on the selected format, the following lists the requirement:

5.1.2.1. Descriptive tag name

- The first letter of each word of the tag name must be capitalized (ex. 350_PM5_1stDryerPressure_PV).
- No spaces, dash or underscore are to be used between words. Caps first letter of each word.
- The use of acronyms is permitted if they are consistent with all other PI tags in a mill. Do use the complete word when possible.
- The use of descriptive tag name is very useful to identify calculation results, totalizers and any other variables that are not related to a control loop or a motor.
- When more than 10 tags need to be created for a similar equipment having the same descriptive name, add an extra digit to the name (ex. Sootblower #1, 670_RB2_SB01_ST to sootblower 78, 670_RB2_SB78_ST).
- When more than 99 tags as before, add an extra digit (ex. Profile tags should be created from 001 to 999).

5.1.2.2. Loop number name

- The tag name must follow ISA Standard S.5 (Instrument Society of Automation) functional identification and loop identification (LIC101 for Level Indicating Controller for loop 101).
- Do not use any separators between the functional descriptor and the loop number (-, _ or space).

- Loop numbers will usually contain an area number before the functional descriptor. This number should be included in the main tag body as it is possible the loop numbers could be duplicated in different areas of the mill.
- All PI tags related to a control loop should have the same prefix and tag name body. The only differentiator will be the suffix.

5.1.2.3. Motor name

- Motor data tag names should use their equipment number as part of the tag name since equipment numbers are unique in each mill.
- A motor tag should have a prefix, a tag name body that includes the equipment number and a suffix.
- As with control loops, all motor PI tags should use the same prefix and tag name body.
- Only the suffix will differentiate between a motor status, a motor load or an alarm.
- It will be very important that the PI tag descriptor field fully describes the service location (ex. PM1 Machine chest pump).

5.1.3. Suffix

The suffix is the last component of a PI tag name. It differentiates the type of data of a PI tag. As mentioned previously all PI tags for a control loop, motor or other equipment must be identical except for the suffix.

The suffix is delimited by either a decimal point (.) or a underscore (_), depending on the mill's standard. The following table shows standard suffix to be used.

SUFFIX	PURPOSE	COMMENTS
AF	Calculation result from PI-AF	Extended descriptor should contain PI-AF database path to the attribute
ALM	Alarm tag	
AVG	Average result	
CALC	Calculation result	
HART-PV	Hart Primary Variable	Hart variables can represent different values depending on the instrument type and version. It is important to add a precise description of the variable in the tag descriptor.
HART-SV	Hart Secondary Variable	Hart variables can represent different values depending on the instrument type and version. It is important to add a precise description of the variable in the tag descriptor.

HART-TV	Hart Tertiary Variable	Hart variables can represent different values depending on the instrument type and version. It is important to add a precise description of the variable in the tag descriptor.
HART-FV	Hart Fourth Variable	Hart variables can represent different values depending on the instrument type and version. It is important to add a precise description of the variable in the tag descriptor.
P	Controller gain	
LAB	Manual data from test data	
LR	Controller Local/Remote	This is reserved for controller having a Boolean status only
MA	Controller Manual/Auto	This is reserved for controller having a Boolean status only
MAN	Manual data entry	
MODE	Controller Mode	
OP or OUT	Controller Output	Some mills use OUT for Controller Output
PV or MV	Process Variables	Some mills use MV for Measured Value
SP	Controller Setpoint	
ST	Status indication	This is used mostly for digital tags like motor run status and valve
I	Controller Integral	
D	Controller Differential	
TF	Controller Filter value	
TOT	Totalizing result	
MAX	Quality specification tag	Quality tags sources are either manual or from PI-AF calculations. If from PI-AF then extended descriptor should contain PI-AF database path to the attribute
MIN	Quality specification tag	Quality tags sources are either manual or from PI-AF calculations. If from PI-AF then extended descriptor should contain PI-AF database path to the attribute
UPPER	Quality specification tag	Quality tags sources are either manual or from PI-AF calculations. If from PI-AF then extended descriptor should contain PI-AF database path to the attribute

LOWER	Quality specification tag	Quality tags sources are either manual or from PI-AF calculations. If from PI-AF then extended descriptor should contain PI-AF database path to the attribute
TARGET	Quality specification tag	Quality tags sources are either manual or from PI-AF calculations. If from PI-AF then extended descriptor should contain PI-AF database path to the attribute

Table 2: Suffix naming standard

6. PI Tag Configuration

PI tag configuration can be divided in five parts: general, archive, classic, security and system. In this section we will describe the parameters of a PI tag.

There are 2 tools that can be used for creating and editing PI tags. These are PI-SMT (System Management Tool) and an Excel add-in called PI-Builder. In the following sections we will be using PI-SMT as it is more visual when working on single tags.

The preferred tool to batch create large number of tags would be the Excel add-in called PI-Builder. The name of the parameters are called the same in both tools.

6.1. General section

The picture below is taken from the PI System Management Tool. It is one of two tools required to create and edit PI tags. The second tool is an Excel Add-In called PI-Builder.

The general section describes the PI tag. We can see attributes that are self-explanatory such as the name, descriptor, PI server name and point type. In this section we will cover all the general tab attributes.

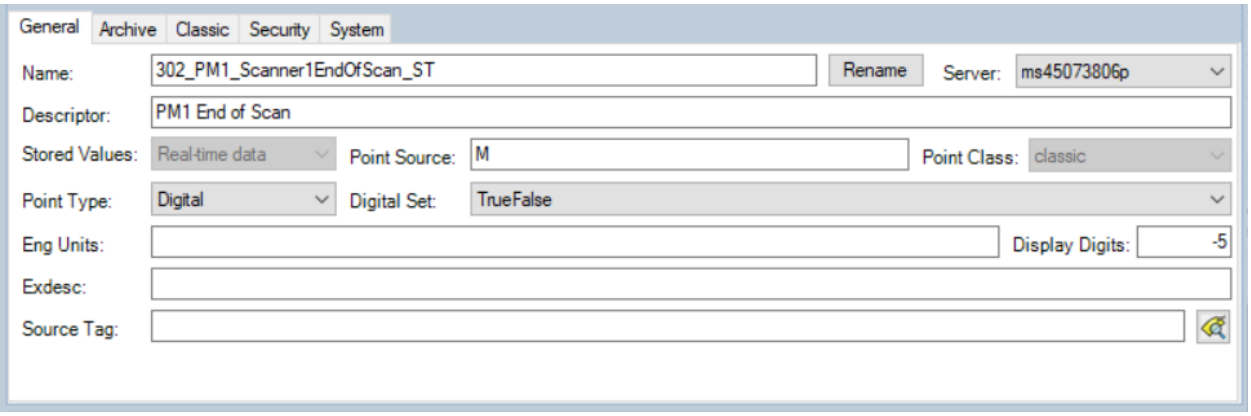


Figure 7: PI Builder General tab

6.1.1. Point Source

The point source identifies the interface node from where the PI tag will be receiving the data. The section 4.2.1 explained how to identify a PI interface. The point source is identified in the "-/PS" parameter of the PI interface.

6.1.2. Point Class

There are 5 classes of PI tags as per the following picture.

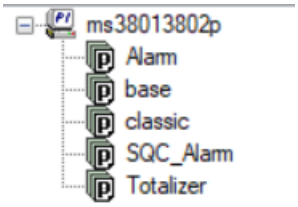


Figure 8: Point Class

The majority of PI tags will be using the Classic class. The alarm class of tag is used only in special applications. The same can be said of the SQC_Alarm class. Totalizer tags are frequently used to calculate averages and totals of sourced PI tags. We will cover totalizer tags is a separate section.

It is not recommended to use base class tags. They are a simpler form of the classic tags and do not offer exception handling of incoming data.

Once created it is not possible to change the Point Class.

6.1.3. Point Type

There are 9 point types as per picture below.

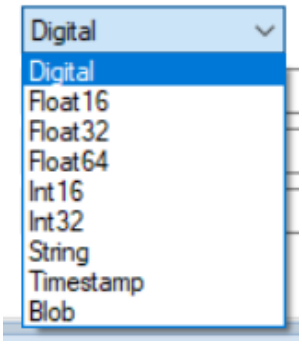


Figure 9: Point Type

- Digital tags are not limited to Boolean only. A digital tag can have multiple states where the value from the interface is converted from a digital number or a string to a digital state.

- Float 16 should never be used. This was left over from very early PI system running on 16 bit computer hardware. The resolution of 16 bit is very poor and are limited by the zero and span of the tag.
- Float 32 is the recommended PI tag type. It offers all the necessary resolution without the drawback of Float 16.
- Float 64 is not recommended. Currently, it offers no additional benefit over Float 32. It will use more disk space and it will take longer to retrieve the data.
- Integer 16 should not be used for the same reason as Float 16
- Integer 32 is recommended when the source data is an integer.
- String will record any string from a PI interface. These are typically used, as an example, for paper machine grade names.
- Timestamp records the timestamp of a variable. This type is not used very often because the PI OPC interface will record the value at its timestamp time. The timestamp will be equal to the scan time unless the PI interface "-TS" parameter is set to "Yes".
- Blob should never be used. This is a legacy point type.

6.1.4. Digital Set

Digital set is only used for digital tags. Otherwise, this parameter is left blank for all other point type.

As mentioned previously, digital tags are not limited to Boolean states. A digital tag reads the data as an integer and then converts the numerical value to a string corresponding the digital set table. The digital set table is configured using the PI System Management Tool.

Each mill has a very comprehensive digital set library. For 2 states digital sets, they are usually named using the first and second states (ex. AutoMan or OnOff or RunStop, etc.).

The picture below shows how to view the digital set library and the individual state name for each state.

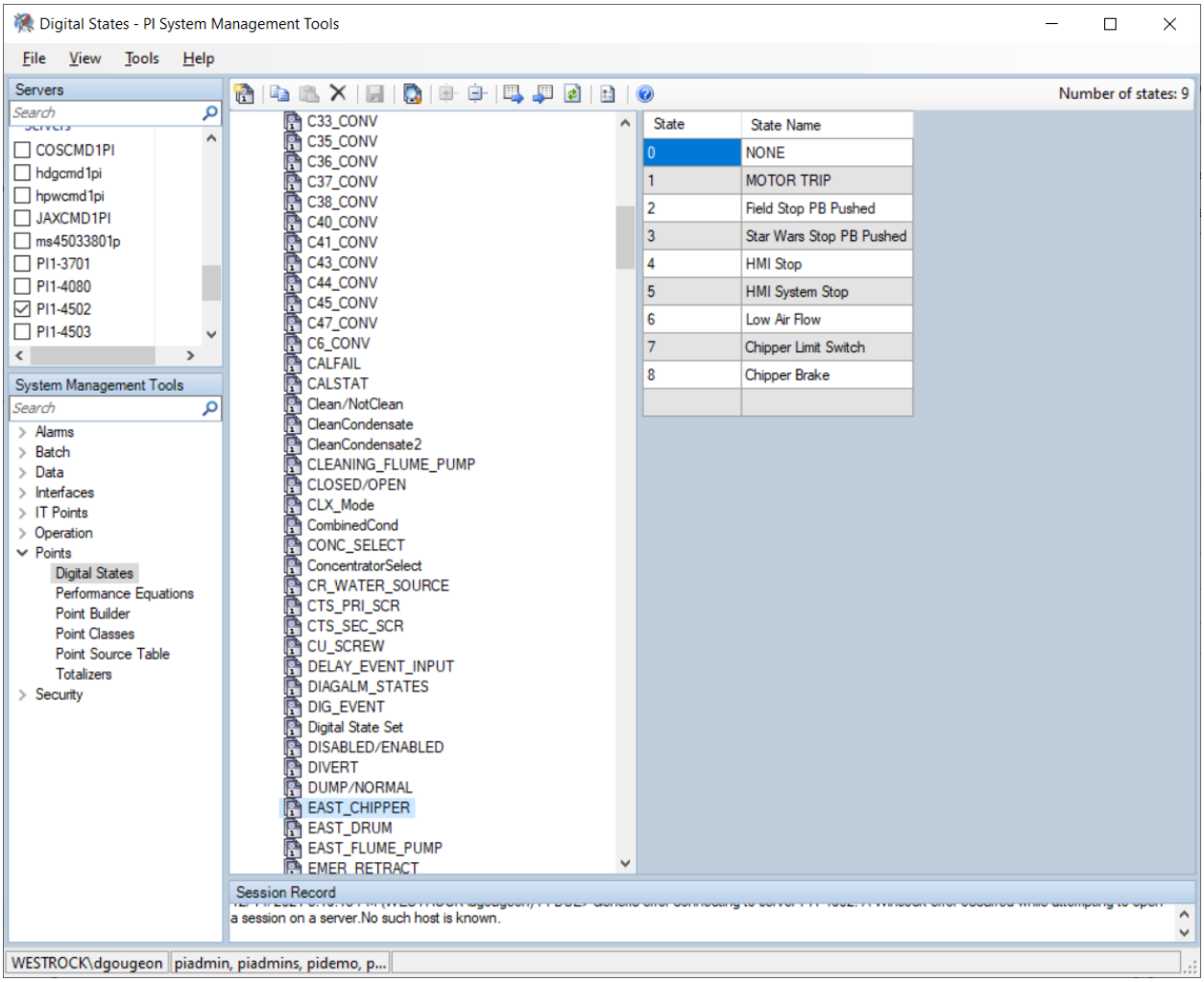


Figure 10: Digital States

It is recommended that for Boolean tags the digital set “False/True” or “True/False” should not be used. It is preferable to assign the proper state to Boolean tags such as “Off/ON” or “Open/Close”.

6.1.5. Engineering Units

Engineering units should exactly match the source such as the DCS units. Digital tags do not require any units. Integer tags units may or may not require units.

Common mistakes in units are levels using pressure units such as inH2O instead of using inches. Another common mistake is using the wrong multiplier. These errors are more common with steam measurement units and conductivity units.

Prefix	Unit Abbreviation	Meaning
giga	G	1000000000
mega	M	1000000

kilo	k	1000
hecto	h	100
deka	da	10
		1
deci	d	1/10
centi	c	1/100
milli	m	1/1,000
micro	µ	1/1,000,000
nano	n	1/1,000,000,000
pico	p	1/1,000,000,000,000

Table 3: Unit Abbreviation Syntax

6.1.6. Display Digits

Display Digits	Result
Sample value	123.456789
-5	123.45
-4	123.5
-3	123
-2	1.2E+2
-1	1E+2
0	123
1	123.5
2	123.46
3	123.457
4	123.4568
5	123.45679

Table 4: Display digits

Display digits default is “-5”. This only applies to floating and integer tags.

This parameter only effects the display of the tag value. It does not change the resolution of the data. The negative sign is a signal to reduce the number of display digits. If the value is larger than display digits then PI will return an exponential number. (ex. Value = 123456.78, display digits = -4 then value = 1.2345E+5

It is possible to configure a positive number of display digits. Using a positive display digit forces a display (ex. ProcessBook display) to show a specified number of significant digits after the decimal point. (ex. Value =123.4567, display digits = 2 then value = 123.46)

6.1.7. Exdesc

Exdesc specifies event points, Long ItemID, Dzero for scaled points, or ItemID to get the time stamp for an output value. It can also contain additional information about the PI tag.

- Event points are tags that do not use a scan class to read the data. Instead, they are triggered by another PI tag. This is especially useful to read profile data on paper machine. The configuration uses the following syntax **event= ' PI_tag_name'**.
- Long ItemID is used only on older version of the PI data archive where the length was limited to 32 characters. The most recent version of the data archive (3.4.370.x) allows 1023 characters. The configuration uses the following syntax **instr=itemID**.
- Dzero is used in combination with scaling parameters. The format to scale is **Dzero=nnnn.nn**.
- The time stamp configuration is used to record in a PI tag the time stamp of an ItemID (another PI tag). This is not necessary if the PI-OPC interface /tf is set to yes (/tf=y).

6.1.8. Source Tag

The Source Tag parameter is used for output tags (points that write data to the point source). this attribute specifies the PI tag that triggers query execution and from which the data is read

6.2. Archive section

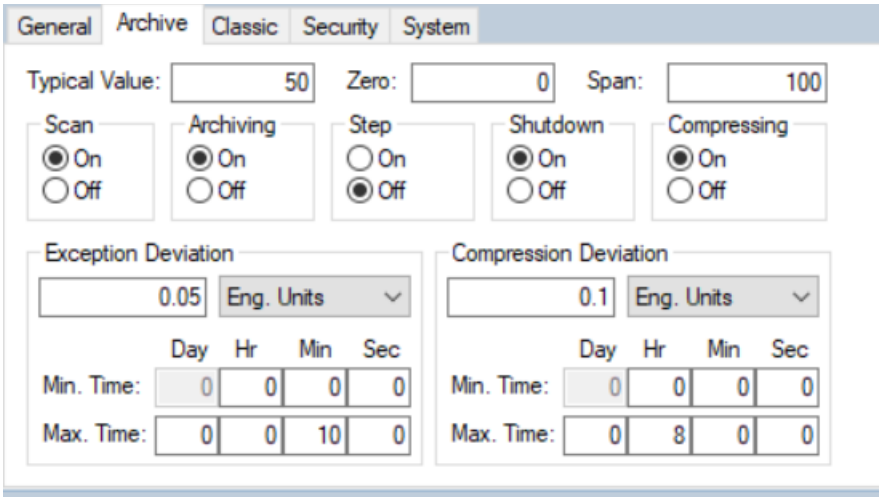


Figure 11: Point Builder Archive tab

The archive section configures the how the PI tag will archive the variables.

6.2.1. Typical Value, zero and span

The zero and span only applies to integer and floating PI tags. They should always match the control system's zero and span. The typical value must be in between zero and span.

For digital tags the zero and span will be grayed out and no entry is required as it will be assigned by the PI server.

6.2.2. Scan

The scan parameter will enable the PI interface to read the value. The Scan = ON is the normal default setting. A tag with scan set to off will return a “Scan Off” error.

6.2.3. Archiving

The archiving parameter enable the PI server to archive the data from the interface. The normal default setting is to be turned ON.

Having the scan on and the archive off the data will be retrieved from the interface. It will show in ProcessBook. However, it will not be recorded in the archive.

6.2.4. Step

The step parameter if turned ON will show data as a step for each archived event. Otherwise, the default configuration should be OFF.

Digital and string tags have the step parameter ON as a default and cannot be changed.

Recommended configurations for Step response are:

- Manual tags should all have Step = ON
- Setpoint PI tags should by default have Step = ON for normal PID loops. Loops capable of cascade mode should have the step parameter set to OFF.
- Calculation that reports on long period of time such as block averages and summary calculations should have the step set to ON.
- Remote operated control valve (not controlled by PID loop output) should be set to step ON.

6.2.5. Shutdown

The shutdown parameter determines if the PI server will write to the PI tag “Shutdown” when the server is rebooted. This parameter is not critical for PI tags from an interface because the interface buffer is configured to write the missing values when the PI server comes back online.

However, having a “Shutdown” value written to a tag performing a calculation may produce a calculation error or worst show an incorrect value. In general, manual tags should have the shutdown parameter set to OFF. Otherwise, the PI server will write a “Shutdown” and every calculation using the manual tag value will show a “Calc Failed”.

6.2.6. Compressing

The compressing parameter determines if the snapshot data from the interface is compressed or not.

The recommended setting is to have **compression always ON**. If the maximum amount of data is required, then Exception Deviation and Compression Deviation should be set to zero.

6.2.7. Exception Settings

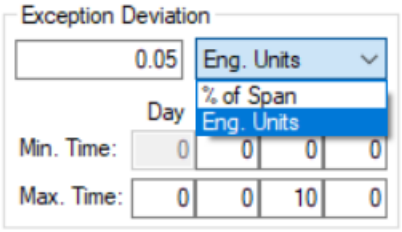


Figure 12: Point Builder Exception Deviation

Exception settings determines how the data is read from the interface. The smaller the value, the more data will be sent to the PI server as Snapshot data. Using PI System Management Tool Point Builder is the easiest method to configure exception handling.

The exception deviation can be configured as a percent of span or in engineering units.

Recommended exception setting:

- The exception deviation should be set at 0.05% of span
- The “Min. Time:” should always be set to zero. In PI Builder Excel add-in, this parameter is called excmin.
- The “Max. Time:” default value is 10 minutes. In PI Builder Excel add-in, this parameter is called excmax and should be set to 600 seconds.

6.2.8. Compression Settings

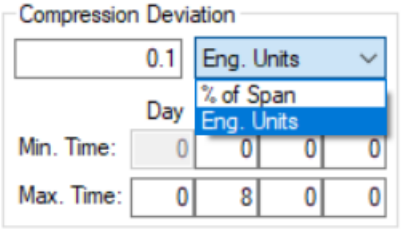


Figure 13: Point Builder Compression Deviation

Compression settings determines how the data is compressed in the PI Server. The smaller the value, the more data will be archived to the PI server as archived events. Using PI System Management Tool Point Builder is the easiest method to configure compression handling.

The compression deviation can be configured as a percent of span or in engineering units.

Recommended compression setting:

- The compression deviation should be set at 0.1% of span
- The “Min. Time:” should always be set to zero. In PI Builder Excel add-in, this parameter is called compmin.

- The “Max. Time:” default value is 8 hours. In PI Builder Excel add-in, this parameter is called compmax and should be set to 28800 seconds.
- Compression deviation should **always be twice the exception deviation**. The objective is to have an archived event every minute for floating PI tags. If there are insufficient archived event, then both the exception and compression deviation should be reduced while keeping the same 2 to 1 ration between compression and exception.

6.3. Classic section

The classic section is used to determine what data is to be access, from what interface, at what frequency, scaling, filtering, and array position.

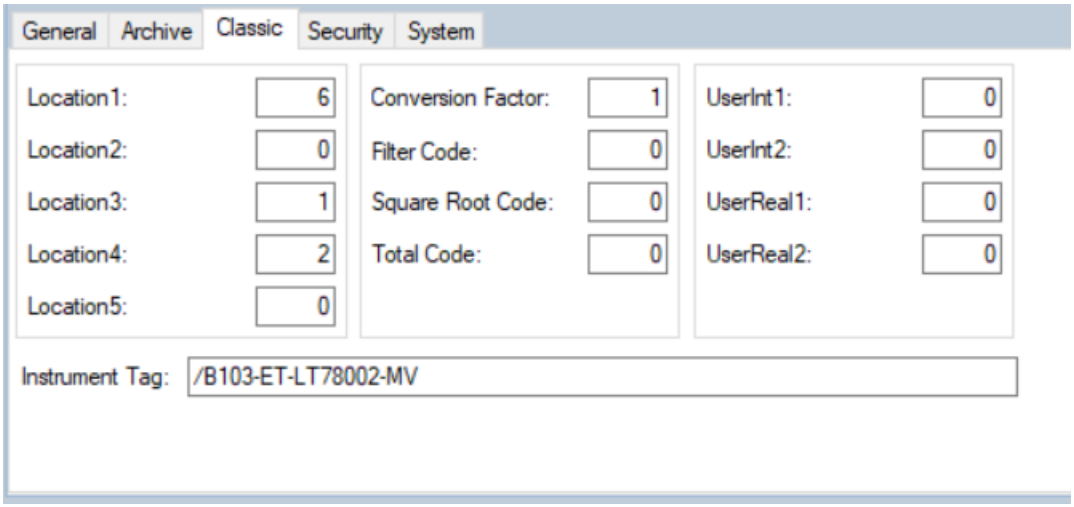


Figure 14: Point Builder Classic tab

6.3.1. Location 1:

Location 1 in combination with PointSource determines what PI interface the PI tag will read (or write) to. The Location 1 corresponds to the PI interface /ID parameter.

6.3.2. Location 2:

The location 2 attribute configures handling of data types for PI Interface for OPC DA.

Valid settings for location2 are as follows:

Value	Description
0	Normal processing; no special handling is used.
1	Read and write value as a string. For digital points, the strings read from the OPC server must match the strings in the digital state set used by the PI point. For integer and real points, the interface requests a string value, and translates it to a number.

2	Read value as a Boolean. Boolean values are either zero or nonzero. For numeric points, any value but 0 (False) is set to -1 (True). Use this option to correctly convert an OPC server Boolean value into the PI System digital state, which prevents the PI point from receiving Bad quality values for a Boolean value that is True.
3	Read value as a four-byte integer. This setting accommodates servers that cannot send the value as a two-byte integer, which is how digital points are normally read.
4	Stores the quality of the item rather than the value.
5	Request real points as VT_R8 items (eight-byte real). By default, the interface requests real points as VT_R4 items (four-byte real). Forfloat32 points (including all PI2 Real points), values that cannot fit in a 32-bit floating-point number lose precision. This setting is included to support servers that do not translate VT_R8 data to VT_R4 data and to permit the use of float32 points where the benefit of greater precision is not worth the overhead of using float64 Points
6	Read time stamps from the OPC server as strings and transform them into seconds. The PI point can be an int or a float. The format of the time stamp string is specified in the interface batch file using the tf parameter.
7	Read time stamps from the OPC server as VT_DATE variables. These values can be translated into time stamp strings or passed to PI Data Archive as a number of seconds, suitable for use in computations. If the value is translated into a string, the format of the time stamp string is specified in the interface batch file using the “/tf” parameter.
8	Directs the OPC server to send the canonical data type. The interface tries to transform the value into the proper data type for the PI point. Use with caution, because the transformation can fail if the source data type is not compatible with the PI point data type, or if the value cannot be represented using the PI point data type.

Table 5: Location 2

Recommended settings for Location 2:

- By default, location 2 should be set to zero for normal processing.
- For Boolean PI tags, location 2 should be set to 2
- In some conditions location 2 can be set to 1 to store the data as a string. This is helpful to record values like paper machine grades.
- All other values for location 2 are not recommended.

6.3.3. Location 3:

Location 3 attribute determines if the PI tag will read or write. Settings 3 and 4 are only used to tags controlling redundant PI server and or interfaces.

Location3	Description
0	Polled or event
1	Advise
2	Output
3	Polled watchdog used with server-level failover
4	Advise watchdog used with server-level failover

Table 6: Location 3

The preferred and recommended value for location 3 is 1 for advise tags. In combination with the interface's /ts = Y parameter, the value will be archived at the same DCS timestamp.

Otherwise, a polled tag will be archived at the time it was read and not at the exact time the value changed. In addition, polled tags create more load on the interface and network traffic back to the PI server.

6.3.4. Location 4:

The location 4 determines what scan class the data will be read.

The table below is the recommended best practice configuration. Each interface may have a different scan class. Therefore, **do not assume** scan classes listed below are correct for your installation.

Scan Class	Sampling interval
1	1 second
2	2 s
3	2 s offset 1 s
4	5 s
	5 s offset 1 s
6	5 s offset 2 s
7	5 s offset 3 s
8	5 s offset 4 s
9	10 s
10	30 s
11	1 minute

Table 7: Location 4

6.3.5. Location 5:

The location 4 is used for deadband processing. The default setting for this attribute is zero.

It is not recommended to use deadband processing. Therefore, location 5 should always be set to zero.

6.3.6. Instrument Tag:

The instrument tag is the actual location name of the variable within the OPC server. The syntax varies with each control system manufacturer.

6.3.7. Scaling attributes:

Scaling incoming data is possible by configuration of "Conversion Factor", "Square Root Code", "DZero" and "Total Code" attributes. It is not necessary to configure a calculation tag for scaling.

The table below shows all the different possible configuration for scaling.

Convers	TotalCode	SquareRoot	Dzero	Operation Input points	Operation Output points
0	0	1	No effect	$(Value)^2$	$(Value)^{0.5}$
		2	No effect	$(Value)^{0.5}$	(Value)2
Nonzero	1	0	Defined	[(Value – Dzero) / Convers] * Span + Zero	[(Value – Zero) / Span] * Convers + Dzero
		1	Defined	[$((Value)^2$ – Dzero) / Convers] * Span + Zero	[$((Value)^{0.5}$ – Zero) / Span] * Convers + Dzero
		2	Defined	[$((Value)^{0.5}$ – Dzero) / Convers] * Span + Zero	[$((Value)^2$ – Zero) / Span] * Convers + Dzero
	2	0	No effect	Value * Convers	Value / Convers
		1	No effect	$(Value)^2$ * Convers	$(Value)^{0.5}$ / Convers
		2	No effect	$(Value)^{0.5}$ * Convers	$(Value)^2$ / Convers
	3	0	Defined	(Value / Convers) – Dzero	(Value + Dzero) * Convers
		1	Defined	$((Value)^2$ / Convers) – Dzero	$((Value)^{0.5}$ + Dzero) * Convers
		2	Defined	$((Value)^{0.5}$ / Convers) – Dzero	$((Value)^2$ + Dzero) * Convers
	4	0	Defined	(Value – Dzero) / Convers	(Value * Convers) + Dzero
		1	Defined	$((Value)^2$ – Dzero)/ Convers	$((Value)^{0.5}$ * Convers) + Dzero
		2	Defined	$((Value)^{0.5}$ – Dzero)/ Convers	$((Value)^2$ * Convers) + Dzero
	5	0	No effect	Value + Convers	Value – Convers

		1	No effect	$(Value)^2$ + Convers	$(Value)^{0.5}$ – Convers
		2	No effect	$(Value)^{0.5}$ + Convers	$(Value)^2$ – Convers
	6	No effect	No effect	Value AND Convers	Value AND Convers
	7	No effect	No effect	Value OR Convers	Value OR Convers
	8	No effect	No effect	Value = Value XOR Convers	Value = Value XOR Convers

Table 7: PI Point calculation matrix

Scaling can be useful where values need to be transformed into proper engineering units. PCL analog tag are often read as integer time a factor of ten. The variable can then be scaled by dividing the incoming value with the proper scaling configuration.

If DZero is required is needed, then it needs to be added to the Exdesc attribute using this syntax: "DZero=nnnn.nn" where "n" is a number.

6.3.8. Filter Code:

The Filtercode indicates the time constant of a first-order filter used to smooth incoming data. It is recommended to not alter incoming data by leaving this code at its default value of 0.

The other options are:

Code	Time Constant (Sec.)
1	10
2	60
3	120
4	600

Table 8: Filter Code

6.3.9. UserInt and UserReal:

Only UserInt1 is used for retrieval of array tags. All other are not used and should be left to the default zero.

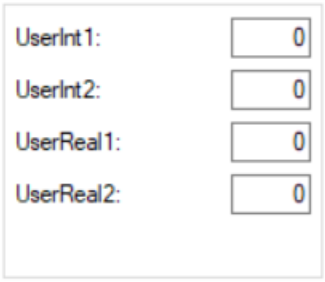


Figure 15: UserInt and UserReal

To configure array tags such as paper machine profiles, a tag for each column of the array is required. All tags must be configured the same except for UserInt1. This attribute represents the column number of the array or the profile slice number.

6.4. Security section

The security section should remain at default and the same for all tags.

Only PI administrators are allowed to edit PI tags. All other users will only have read access to the data.

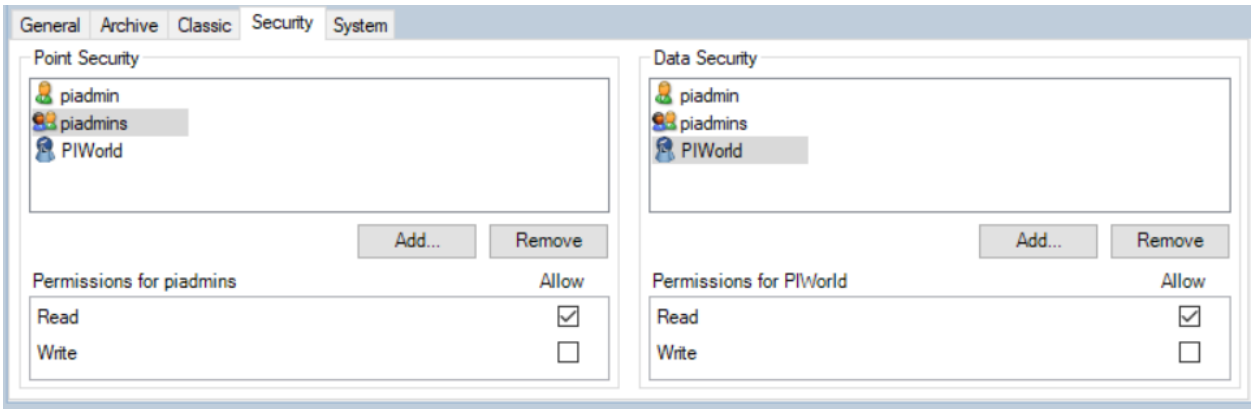


Figure 16: PI Point Security tab

6.5. System section

There is input to be done in the system section. It only serves as information regarding the PI tag such as who and when it was created as well as when it was last changed.

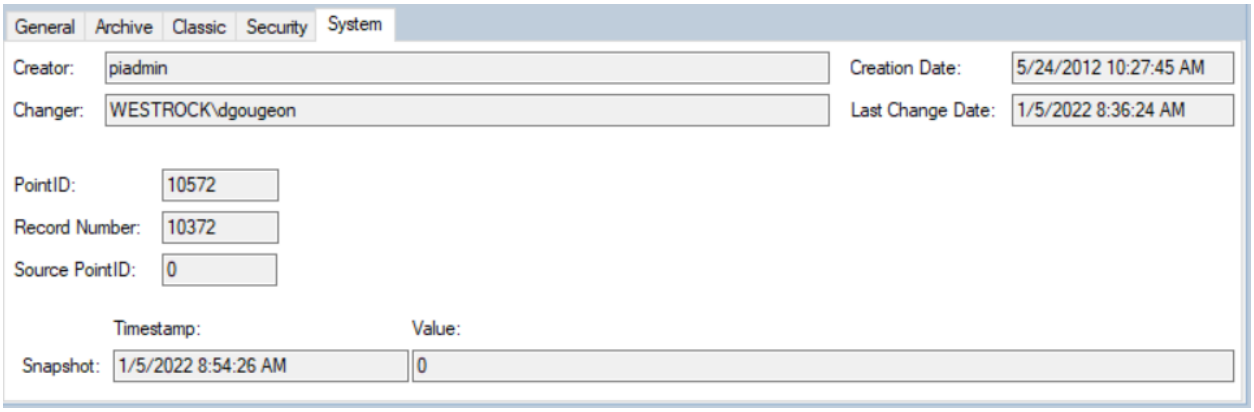


Figure 18: Point Builder System tab

7. PI Tag Best Practices

The following section will provide examples of PI tags for control loops and motors.

7.1. Control Loops

The recommended control loop tags should include the following:

- PV for the process variable
- OP for the controller output
- SP for the controller setpoint
- MODE for the controller mode
- P for the controller gain
- I for the controller integral
- D for the controller differential
- TF for the DCS input filter
- FF for the controller feedforward value if the loop is configured to have feed forward
- HART-PV, HART-SV, HART-TV and HART-FV for smart valve positioner feedback.

The body of the tag must be identical to keep all the control loop PI tags together.

The HART 4 primary variables may have different functions depending on the manufacturer, model of the smart positioner and firmware version. It will be critical to properly identify in the tag description the function of the different variables.

Depending on the manufacturer of the DCS it may be possible to only have Boolean status for the controller mode. In general, these control systems will all have a MA for Manual/Auto mode. If the control loop is configured as a cascade or remote-control loop, then a second PI tag will need to be created. This is where LR for Local/Remote is required. Consequently, a MODE calculated tag will need to be configured in PI-AF.

7.1.1. Tag naming

All the tags for a control loop should have the same name with the appropriate suffix. This is regardless of the instrumenttag.

Example:

Tag name:	421-LIC-1010-01.PV	Instrumenttag:	AO04.421-LIC-1010-01_me
	421-LIC-1010-01.OUT		AO04.421-LIC-1010-01_pos
	421-LIC-1010-01.SP		AO04.421-LIC-1010-01_spa
	421-LIC-1010-01_HART-SV.PV		AO01.421-LIC-1010-01M_SV_av

The examples above shows that the PI tag name follows the naming convention and is totally independent of the instrumenttag parameter.

7.1.2. Tag scan class (Location 4)

The scan class determines the frequency of the interface reading the OPC server. The Location 3 determines the scan class and NOT the read interval. The read interval is determined by the interface configuration and the "/F" parameters.

As a best practice, it is recommended to configure control loop PI tags for a 5 second frequency. It may be necessary to have faster sampling on critical fast-moving loops to optimize control loop performance.

For large count of tags on a single interface, it is recommended to distribute the PI tags between scan class 4 thru 8.

7.1.3. Location 3

Location 2 determines if the tag is to be polled or advised. It is recommended to configure all PI tags to be in advise mode (Location 3 = 1)

7.1.4. Zero and Span

The zero and span should always match the DCS configuration.

7.1.5. Exception and Compression

The recommended exception setting should always be ½ of compression or less. For good performance of the PI system compression algorithm, it is necessary to have enough values also called snapshots from the interface.

The recommended exception should be 0.05% of span and the compression should be 0.1% of span. These settings are only a good starting point. The objective is to have 1440 archives events per day up to a maximum of 10,000 events per day. It may be necessary to adjust both exception and compression for good data quality.

7.1.6. Setpoint tags

Control loop setpoint should have step = 1, exception = 0 and compression = 0.

The objective is to capture all setpoint changes regardless of the span. The only exception would be control loops that are configured to operate in “Cascade” or “Remote” setpoint. For these loops it is recommended to set step = 0, exception and compression to half of PV exception and compression.

7.1.7. Shutdown

The shutdown parameter should be set to “Off” for all tags sourced from an interface. Only manual and calculated tags should have shutdown set to “On”.

7.2. Motor tags and other digital tags

Motor tags should include the following:

- Motor run status
- Motor load in either % full load or in amps
- Variable speed motors should have speed feedback
- Any alarm messages

As with control loop PI tags, all motor PI tags should have the same prefix and body name. Only the suffix will differentiate each tag.

In general motor load tags best practice for exception and compression will be 0.125% and 0.25% of span. Scan class should be set to 5 seconds with advise mode (location 3 = 1).

The shutdown parameter should be set to “OFF” (Shutdown = 0).

7.3. Manual tags

Manual tags can be organized in 2 categories. Some PI manual tags are used in calculations that only change infrequently such as the cost of a chemical. Others are tags having source such as lab test.

The only parameter to be concerned with would be the “Shutdown”. The safe option is to have shutdown set to “OFF”. However, for lab test data it may be better to have shutdown set to “ON” because it may be misleading if an old value is still showing.

7.4. PI AF tags

PI AF tags should not be created from AF. We need to have AF tags as “Classic” and not “Base” tag type. Secondly the tag naming convention should follow the tag naming convention and have the same suffix and tag body unless this is a calculation that is not related to a control loop or motor (ex. Production summary calculation attribute).

7.4.1. Exception and Compression

As a general rule, exception and compression should be set to zero.

7.4.2. Step response

The step response should be applied for elapse time summary calculations. Otherwise, step should be set to zero.

7.4.3. Shutdown

The shutdown parameter for AF tags should be set to “ON”.

7.4.4. Extended descriptor and Point source

The point source for AF tags should always be set to “AF” to indicate the values are being written by an AF database.

It is very important for documentation and troubleshooting purpose to write the full path of the AF attribute writing to the AF tags In the Exdec. This path should include the database name, element, child elements and attribute.

As an example, the Exdec should show something like this “\\Fernandina\4502\Fernandina Beach\Chemical Recovery\Evaporators\6 EVAP\Concentrators\C1A Concentrator\Black Liquor\Feed Flow Control”.